WORKED SOLUTION

STEP III 2025, Question 12

Probability and Statistics • Recursive Random Variables

STEP III 2025, Question 12

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

The key idea

Part (i) is just a double sum written two ways: both sides add up $\mathrm{f}(r)\,\mathrm{g}(s)$ over the same set of pairs (r,s), one counting row by row and the other column by column. In part (ii), conditioning on the previous value X_{n-1} gives each probability $\mathrm{P}(X_n = r)$ as exactly the kind of sum that appears on the left of (i). Swapping the order of summation then turns $\mathrm{E}(X_n)$ into something we can recognise in terms of $\mathrm{E}(X_{n-1})$.

Solution

Part (i)

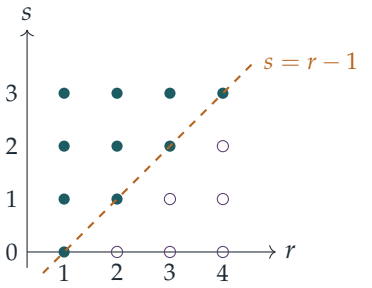
Expanding the brackets, the left-hand side is

$$\sum_{r=1}^{m+1}\sum_{s=r-1}^{m}\mathrm{f}(r)\,\mathrm{g}(s),$$

a sum of $\mathrm{f}(r)\,\mathrm{g}(s)$ over all integer pairs (r,s) with

$$1 \le r \le m+1 \quad \text{and} \quad r-1 \le s \le m.$$

We describe the same set of pairs by fixing s first. The conditions force $0 \le s \le m$ (since $s \ge r-1 \ge 0$), and for a given such s the allowed r are those with $1 \le r$ and $r \le s+1$; note $s+1 \le m+1$, so the bound $r \le m+1$ is then automatic. Conversely any pair with $0 \le s \le m$ and $1 \le r \le s+1$ satisfies the original conditions. The picture below shows the triangle of pairs for $m = 3$.



Summing by columns (fixed r) gives the left-hand side; summing by rows (fixed s) gives

$$\sum_{s=0}^{m}\sum_{r=1}^{s+1} f(r)\,g(s) = \sum_{s=0}^{m}\Big(g(s)\sum_{r=1}^{s+1} f(r)\Big),$$

which is the right-hand side, as required. Since the sums are finite, reordering them is always allowed.

Part (ii)(a)

X_1 is equally likely to be 0 or 1, so $\mathrm{E}(X_1) = \frac{1}{2}$.

For X_2, condition on X_1. If $X_1 = 0$, then X_2 is uniform on $\{0,1\}$; if $X_1 = 1$, then X_2 is uniform on $\{0,1,2\}$. By the law of total probability,

$$\begin{aligned}
\mathrm{P}(X_2 = 0) &= \tfrac{1}{2}\cdot\tfrac{1}{2} + \tfrac{1}{2}\cdot\tfrac{1}{3} = \tfrac{5}{12},\\
\mathrm{P}(X_2 = 1) &= \tfrac{1}{2}\cdot\tfrac{1}{2} + \tfrac{1}{2}\cdot\tfrac{1}{3} = \tfrac{5}{12},\\
\mathrm{P}(X_2 = 2) &= \tfrac{1}{2}\cdot 0 + \tfrac{1}{2}\cdot\tfrac{1}{3} = \tfrac{1}{6},
\end{aligned}$$

as required (and the three add to 1, a useful check). Hence

$$\mathrm{E}(X_2) = 0\cdot\tfrac{5}{12} + 1\cdot\tfrac{5}{12} + 2\cdot\tfrac{1}{6} = \tfrac{3}{4}.$$

Part (ii)(b)

By induction X_{n-1} takes values in $\{0,1,\ldots,n-1\}$ (each step raises the maximum possible value by one). Given $X_{n-1} = s$, X_n is uniform on the $s+2$ values $0,1,\ldots,s+1$, so

$$\mathrm{P}(X_n = r \mid X_{n-1} = s) = \begin{cases} \dfrac{1}{s+2}, & 0 \le r \le s+1,\\ 0, & r > s+1. \end{cases}$$

The value $r = 0$ is always possible, so the law of total probability gives

$$\mathrm{P}(X_n = 0) = \sum_{s=0}^{n-1} \frac{\mathrm{P}(X_{n-1} = s)}{s+2}.$$

For $1 \le r \le n$, only those s with $s+1 \ge r$, that is $s \ge r-1$, contribute:

$$\mathrm{P}(X_n = r) = \sum_{s=r-1}^{n-1} \frac{\mathrm{P}(X_{n-1} = s)}{s+2}, \qquad r = 1,2,\ldots,n.$$

(For $r = 1$ this is the same sum as for $r = 0$, so $\mathrm{P}(X_n = 0) = \mathrm{P}(X_n = 1)$ for every $n \geq 1$, matching the two values $\frac{5}{12}$ in (a).)

Part (ii)(c)

Write $p_s = \mathrm{P}(X_{n-1} = s)$. The $r = 0$ term contributes nothing to the expectation, so by (b)

$$\mathrm{E}(X_n) = \sum_{r=1}^{n} r\,\mathrm{P}(X_n = r) = \sum_{r=1}^{n} \Big(r \sum_{s=r-1}^{n-1} \frac{p_s}{s+2} \Big).$$

? **Ask yourself** How does part (i) apply here?

This is the left-hand side of (i) with $m = n - 1$ (allowed, as $n \geq 1$), $\mathrm{f}(r) = r$ and $\mathrm{g}(s) = p_s/(s+2)$. Swapping the order of summation,

$$\begin{aligned} \mathrm{E}(X_n) &= \sum_{s=0}^{n-1} \Big(\frac{p_s}{s+2} \sum_{r=1}^{s+1} r \Big) = \sum_{s=0}^{n-1} \frac{p_s}{s+2} \cdot \frac{(s+1)(s+2)}{2} \\ &= \frac{1}{2} \sum_{s=0}^{n-1} (s+1)\, p_s = \frac{1}{2} \Big(\sum_{s=0}^{n-1} p_s + \sum_{s=0}^{n-1} s\, p_s \Big) \\ &= \tfrac{1}{2} \big(1 + \mathrm{E}(X_{n-1}) \big), \end{aligned}$$

using $\sum p_s = 1$. This is the required result.

Intuitively: given $X_{n-1} = s$, the mean of a uniform variable on $\{0, \ldots, s+1\}$ is $\frac{1}{2}(s+1)$, so this is the tower law in disguise.

To solve the recurrence, write $e_n = \mathrm{E}(X_n)$. Subtracting 1 from both sides,

$$e_n - 1 = \tfrac{1}{2}\,(e_{n-1} - 1),$$

so $e_n - 1$ halves at each step. Since $e_0 = 0$, we get $e_n - 1 = -\big(\frac{1}{2}\big)^n$, that is

$$\mathrm{E}(X_n) = 1 - \frac{1}{2^n}, \qquad n = 1, 2, \ldots.$$

Checks

The formula gives $\mathrm{E}(X_1) = \frac{1}{2}$ and $\mathrm{E}(X_2) = \frac{3}{4}$, agreeing with (a). Computing the exact distribution of X_n step by step up to $n = 15$ confirms both the expressions in (b) and $\mathrm{E}(X_n) = 1 - 2^{-n}$; for example $\mathrm{E}(X_5) = \frac{31}{32} \approx 0.969$, and a computer simulation of 200 000 runs gave 0.967. As $n \to \infty$ the mean tends to 1, which is plausible: each step pulls the mean halfway towards 1.

Common slips

- Starting the sum for $\mathrm{P}(X_n = r)$ at $s = r$ rather than $s = r - 1$: the value r is reachable from $s = r - 1$, since X_n can go up to $X_{n-1} + 1$.

- Getting the limits wrong when swapping the order in (c): after the swap, r runs from 1 to $s+1$, giving $\frac{1}{2}(s+1)(s+2)$, and the factor $s+2$ cancels neatly.
- Solving the recurrence with e_1 as starting value but writing $1-2^{-(n-1)}$; check against $e_1 = \frac{1}{2}$.

Beyond STEP

The process here is a Markov chain: the distribution of X_n depends only on X_{n-1}, and (b) is one step of multiplying a probability vector by a transition matrix. The shortcut in (c) is the law of total expectation, $\mathrm{E}(X_n) = \mathrm{E}\big(\mathrm{E}(X_n \mid X_{n-1})\big)$, which reduces many such problems to a linear recurrence. Both ideas are developed in Grimmett and Welsh, *Probability: An Introduction*.

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